

System Design Blueprint & Latency Numbers Handbook

Architecture guide for distributed systems and high-level design interviews

1. Step-by-Step 45-Minute Interview Strategy

00-05 min: Scope Requirements (Functional, Non-Functional, Read/Write ratio).

05-10 min: Capacity Estimation (Storage over 5 years, QPS, Bandwidth).

10-25 min: High-Level Architecture (API, Clients, LB, App Servers, DB, Cache).

25-40 min: Deep Dives (Sharding key, Replication, Concurrency, Failure modes).

40-45 min: Bottlenecks & Wrap-Up (SPOF, Monitoring, Metrics, Cost).

2. Latency Numbers Every Engineer Must Know

L1 cache reference: 0.5 ns | Branch mispredict: 5 ns

L2 cache reference: 7 ns | Mutex lock/unlock: 25 ns

Main memory reference: 100 ns

Read 1 MB sequentially from memory: 250,000 ns (250 us)

SSD random read: 150 us | Read 1 MB sequentially from SSD: 1,000 us (1 ms)

Round trip within same datacenter: 500 us (0.5 ms)

Send packet California to Netherlands: 150,000 us (150 ms)

3. Storage & Database Selection

Relational (PostgreSQL, MySQL): ACID transactions, structured schemas, complex joins.

NoSQL Document (MongoDB, DynamoDB): Unstructured JSON, dynamic schema, high write scale.

Wide-Column (Cassandra, ScyllaDB): Massive write throughput, time-series, append-only logs.

In-Memory (Redis): Low latency (<1ms) caching, pub/sub, rate-limiting, session store.